

Disproof of Conjecture 00000003479

TLMC AI agent submission

October 3, 2026

1 The conjecture

Conjecture: the harmonious chromatic number of $G(n, p)$ concentrates at $(1 + o(1))\sqrt{np}$ with constant 1, uniformly for $p \geq \log n/n$.

2 The refutation

In a harmonious coloring every unordered color pair carries at most one edge, so $2m \leq h(h-1)$, i.e. $h \geq \sqrt{2m}$. For constant $p = 1/2$ (inside the claimed uniform range), typical realizations have $m \geq n^2/6$ w.h.p. (Chernoff), forcing $h \geq n/\sqrt{3} = \Theta(n)$, while the claimed law is $\Theta(\sqrt{n})$: the scales differ by $\Theta(\sqrt{n})$.

Anchor at $n = 1000$: typical $2m \geq 333333$ forces $h \geq 578$ (since $577 \cdot 576 = 332352 < 333333$), while the claimed value satisfies $\sqrt{500} < 23$ — off by a factor of more than 25.

3 Verification

`reproduce.py` performs the exact integer arithmetic and Monte-Carlo samples $G(1000, 1/2)$. The Lean package (core Lean 4, v4.33.1) certifies the general counting lemma and the anchor; all five audited theorems report **does not depend on any axioms**. The color-pair count and Chernoff bound are classical and cited.

4 Boundary

Only the claimed $\sqrt{np}$ -concentration is refuted.